

Development of an Image Processing System for Plant Phenotyping in Low-cost Automated
Greenhouse Environments

by

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Abstract

Human population growth continues to strain the global food supply, and agricultural production needs to increase by 70% by 2050 to meet this demand. As a result, the development of cultivars that can produce the most yield and are resistant to various adverse abiotic and biotic stressors has become increasingly important. Plant phenotyping methods are employed to investigate gene-environment interactions in the development of such cultivars, and over the past decade, automated image-processing and analysis techniques in plant phenotyping have assisted in these investigations. Despite that, automated image-based tasks can become costly. Consequently, there is a need for the development of low-cost, automated image-based plant phenotyping systems in greenhouse environments. This study develops an open-source image-processing-based plant phenotyping component, which helps to detect various factors of plant health, stress and growth in order to support low-cost automated greenhouses.

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Introduction

The rapid growth of the human population has created a strain on international food security. Consequently, crop production must increase by 70% by 2050 to meet this demand [1]. Over the past decade, methods in image-processing-based plant phenotyping have been employed to investigate plant gene-environment interactions in greenhouse environments to develop cultivars that exhibit optimal yield and resistance to abiotic and biotic stressors. Despite these advances, current solutions are non-generalizable to a broad range of plant species and can be costly [2, 3, 4]. Thus, there is a critical need for affordable, low-cost plant phenotyping image processing tools in automated greenhouses. This study develops an open-source image-processing-based plant phenotyping component, which helps to detect various factors of plant health, stress and growth in order to support low-cost automated greenhouses. The low-cost imaging component developed utilises low-complexity and efficient image segmentation techniques suitable for resource-constrained hardware, leveraging open-source software and publicly available labelled plant image datasets. Additionally, our component is extensible and flexible, making it more accessible and generalisable to other plant environment contexts. Following this introduction, we review the literature on image processing in greenhouse environments, paying particular emphasis to plant phenotyping and its applications with image processing in automated, low-cost greenhouse environments. Thereafter, we discuss our methodology for developing the low-cost image processing tool, detailing our automated image processing and analysis pipelines. Lastly, we discuss the effectiveness and cost-effectiveness of the developed component using publically available plant image datasets.

Literature Review

2.1 Introduction

This literature review provides a foundational understanding of image processing in plant phenotyping and directions for developing or investigating solutions for low-cost image-processing phenotyping tools in automated greenhouses. The focus is on the existing image processing solutions applied in low-cost automated greenhouse environments. We begin by examining the fundamental concepts of plant phenotyping and the various traits that are amenable to image processing for predicting plant health, growth, and stress. Specifically, we note that leaf phenotypic traits are a robust proxy indicator of physiological and metabolic factors affecting plant health [5].

Thereafter, aspects of image processing in plant phenotyping are discussed. Notably, image processing and acquisition, as well as various image sensor technologies and segmentation methods, are reviewed in the context of plant phenotyping. In particular, hyperspectral and RGB imaging are the most common acquisition methods, and segmentation techniques critically influence the overall accuracy, precision, and quality of phenotypical trait analysis [6, 7].

Lastly, the current status, challenges, and potential innovations of low-cost automated image processing in greenhouse environments are discussed. In this vein, the literature indicates a lack of affordable and generalizable image processing tools that can analyse and process large amounts of image data on inexpensive, resource-constrained hardware and sensors [8, 4].

2.2 Plant Phenotyping

Plant breeding is defined as the process of inducing desirable inheritable traits in plants through controlled human action [9], which is important in solving the problem of providing an adequate global food supply. The rapid growth of the human population has created a strain on international food security. It

is estimated that crop production must increase by 70% by 2050 [1] to meet this demand. Additionally, abiotic and biotic stressors limit cultivar biomass and yield [10]. Therefore, plant breeders need effective techniques to produce germplasm for cultivars that maximize yield and biomass, in various adverse environmental conditions. In the last few decades, image processing techniques in plant phenotyping have been an effective tool for plant breeding [11].

2.2.1 What Is Plant Phenotyping?

Plant phenotyping is the evaluation of phenotypes in plants [12], where phenotypes are primitive or complex traits of an organism that can be observed or quantified and evaluated numerically [13]. Phenotypes can alternatively be seen as a combination of all morphological, physiological, chemical, and behavioural aspects that represent the individual plant [14, 15, 3].

2.2.2 Plant Phenotyping As Method For Improving Plant Traits

Quantitative data produced by plant phenotyping is most useful in analyzing plant-environment interactions in plant breeding [12] for the development of germplasm with improved plant traits [3]. The data produced by phenotyping is used as a proxy for assessing a plant’s performance in relation to its environment. For example, stressors or disturbances of plants are induced by environmental conditions that organisms are placed in, and these stressors induce symptoms which can be detected and evaluated using phenotyping. This indicates phenotypical traits of a plant have a complex dynamic feedback loop with their environment [16]. Therefore phenotypical traits serve as proxies in determining whether certain genotypes have favourable traits in particular environmental conditions.

2.2.3 Image-Obtainable Phenotypical Traits As Indicators Of Plant Health, Growth, And Stress

A phenotypical trait is classifiable as morphological, physiological, or temporal. Additionally, the literature indicates morphological and physiological phenotypical traits are classifiable as holistic-based or component-based [17]. Holistic properties consider the plant as a whole, whereas component-based features look at organs like the organism’s fruit, flower, leaves, stem, etc.

2.2.3.1 Leaf-Derived Phenotypical Traits: Robust Proxy Indicators of Plant Metabolic and Physiological Status

Leaf-derived traits are one of the most commonly used plant status indicators because leaves rapidly respond to environmental stimuli and robustly indicate various factors that determine plant health, growth and stress [5]. Traits of leaves can monitor and predict a host of plant health, growth and stress factors. For

example, leaf area predicts plant productivity because reduced leaf area results in decreased chlorophyll content and less photosynthesis. Therefore, leaf area is a good proxy measure for evaluating plant growth and has been used to monitor and estimate crop yield and health [18]. Various other leaf morphological traits exist in plant phenotyping. These include leaf number and inclination angle for monitoring plant growth and development [19, 5], and leaf thickness for measuring abiotic stressors like water scarcity [20]. Additionally, physiological traits of leaves exist that indicate various factors of plant status like stomatal conductance for the detection of biotic stressors like pathogens [21], leaf water content for drought stress [22].

2.2.3.2 Alternative Phenotypical Traits

Although leaves play an important role in evaluating plant growth, health and resilience to stress, these evaluations cannot be scaled to the whole plant [5]. Rather, other parts of the plant should be phenotyped to understand the processes governing plant status comprehensively. The literature is rife with such examples that study traits of roots [23], stems [24], seeds [25], and fruit [26].

2.3 Image-Based Plant Phenotyping

Traditional plant phenotyping methods are destructive, prone to human error, time-consuming, and challenging to perform from start to end of a plant’s ontogenesis [2, 5, 27]. Recent advances in image sensing, processing, and data management have produced high-throughput image-based plant phenotyping systems, which are amenable to automation and can analyse large samples of data precisely in a very short period. Reducing the need for potentially erroneous, time-consuming, expensive human labour in plant phenotyping [17].

These Modern high-throughput image-based plant phenotyping systems are examples of cyberinfrastructure (CI) systems [26], which are research environments that support data acquisition, storage, management and computing and processing services distributed over some network [28]. For image-based plant phenotyping, this means that image acquisition and processing techniques are vital.

2.3.1 Acquisition Techniques

Image processing techniques in plant phenotyping cannot be applied until suitable images are acquired. Image sensor advancements have provided the foundations for detecting various information from plants using electromagnetic radiation. Plant tissues interact with this radiation by absorbing, reflecting, emitting, transmitting, and fluorescing light. Different ranges of wavelengths correspond to different biochemical or physiological aspects of plant tissue,

which are not visible to the naked eye [3]. Thus, the choice of imaging technique for data acquisition affects the types of phenotypical parameters that can be measured.

2.3.1.1 Common Image Sensor Techniques In Plant Phenotyping

Current imaging techniques are visible, fluorescence, thermal, tomographic, and hyperspectral imaging [29, 3], where the most commonly used techniques are visible, and hyperspectral imaging due to their affordability, robustness and flexibility in measuring a wide range of phenotypic parameters. The most common imaging technique is visible-imaging. Typically affordable, and high-resolution RGB cameras are used in visible imaging. These devices are sensitive to the visible spectral range and have been used to measure various phenotypic parameters like shoot biomass, yield and leaf morphology, and salinity tolerance [30, 31]. Hyperspectral imaging is another widely used in plant phenotyping. In this technique, sensors capture multiple spectral ranges, enabling detailed measurements for various phenotypic parameters. Notable applications of hyperspectral imaging include estimating nutrient content and detecting plant responses to abiotic or biotic stressors [32]. The other imaging techniques are not as widely used as visible or hyperspectral imaging; however, they are also effective in measuring plant phenotypic parameters. For instance, thermal imaging in plant water stress relations [3, 21], fluorescence imaging in used plant pathogenic symptom monitoring [33, 34], and tomography in estimating plant organ water content [35].

2.3.2 Processing Techniques

2.3.2.1 The importance of Image Segmentation in Plant Phenotyping

There is a lack of generalised or standardised image-processing techniques in plant phenotyping; hence, bespoke image-processing pipelines are designed for specific plant species or genotypes like *Arabidopsis* [3, 36]. Therefore, various phenotyping systems' image processing and analysis pipelines can vastly differ; however, as mentioned earlier in the review of plant phenotyping traits, various morphological traits are derived from specific organs or parts of the plant. Furthermore, segmentation methods directly affect the accuracy of more complex tasks like object detection, 3D reconstruction, and classification. Hence, the validity and preciseness of the derived phenotypical measurements and analysis depend on the image segmentation method selected [6, 7].

2.3.2.2 Commonly Used Image Segmentation Techniques

In plant phenotyping, appropriate selection of segmentation techniques is vital when considering the type of phenotypic parameter to be measured. Segmentation methods in plant phenotyping are typically classifiable as traditional or learning-based [6]. Traditional segmentation methods in plant phenotyping

typically target holistic phenotypical traits like plant height and include frame differencing [37], colour-based [38], edge detection [39], spectral band difference-based [40], shape modelling-based [41], and graph-based segmentation [42]. In contrast to traditional techniques, learning-based methods are more suitable for targeting component-based traits of specific organs or parts of a plant [6]. Many learning-based methods use deep learning and machine-learning mechanisms to handle segmentation [43, 44].

2.4 Low-cost Automated Image-Based Plant Phenotyping In Greenhouses

There is a growing need for low-cost, flexible, modifiable, and extendable automated image processing systems for plant phenotyping applications. Many automated phenotypical image processing systems in greenhouses employ various advanced technologies, such as robotics [45], automated conveyor belts, and benchtop configurations [46, 47], and advanced image sensors in high-throughput phenotyping systems [48]. However, these solutions are not accessible to all plant growers because they tend to be costly [49, 50, 4], and many systems are patented by specialized companies, rendering the hardware and software inflexible and unmodifiable [51]. Furthermore, existing low-cost phenotyping systems in greenhouses have bespoke image analysis and processing pipelines, which prevent them from being generalized and used in broader contexts [2, 3]. Therefore, there is a need for low-cost, flexible, modifiable, generalizable, and extendable automated image processing systems for plant phenotyping applications.

2.4.1 Open-source Software, Systems and Affordable Hardware As Cost Reducers In Image-Based Plant Phenotyping

Recent studies in low-cost automated image-based plant phenotyping show increased development of open-source phenotyping systems and the use of readily accessible commercial off-the-shelf hardware [52, 4] to reduce the costs of image-based phenotyping in greenhouse environments. Cheap low-cost compute sensors called Raspberry PIs have been used with affordable RGB and hyperspectral sensors to monitor phenotypic traits in greenhouse environments [50]. Additionally, there has been an increase in open-source image-processing plant phenotyping systems like Phenobox [51].

2.4.2 Challenges In Automated Image-Based Plant Phenotyping

2.4.2.1 Large Image Sensor Data Throughput Necessitates Robust Image Processing Tools

Effective choices in image sensors are crucial for developing low-cost, automated image-based phenotyping systems [5]. Modern image sensors in image-based phenotyping generate massive amounts of data [8], necessitating the development of efficient image processing pipelines in these contexts to handle large datasets in the shortest possible time while also delivering precise and accurate phenotypical analysis [4].

2.4.2.2 The Effects of Image Sensor Choice On Phenotypical Analysis And Processing

The choice of sensor has a non-trivial influence on the image analysis and processing techniques used in plant phenotyping and additionally also determines the breadth of traits that can be analysed [3]. For example, hyperspectral imaging is practical for studying plant metabolic status; however, the sensors required generate Terabytes of data when monitoring multiple plants [53]. Consequently, in low-cost settings, hyperspectral imaging of component-based phenotypical traits becomes infeasible because learning-based image techniques, which are most suitable for this purpose, are costly, time-consuming, and impractical when large amounts of phenotypical data are processed [53]. Additionally, low-cost hardware has constrained bandwidth, storage, and compute power. Therefore, remote cloud image processing and analysis are required [54], which can be expensive [4].

2.4.2.3 Solutions, and Limitations For Low-cost Large Data Throughput Image Processing Systems

In low-cost settings, commonly used strategies to manage the issues of large-scale image analysis and processing of phenotypic data include low-complexity traditional segmentation methods [4], the use of more affordable and straightforward imaging sensing technologies in low-cost off-the-shelf hardware [50], and image compression techniques [55]. However, each of these strategies comes with trade-offs in terms of precision, accuracy, and phenotypic analysis capabilities. Low-complexity traditional segmentation methods struggle to handle component-based phenotyping of specific organs in plants while requiring fewer computational resources [6]. RGB sensors are affordable and generate less raw data than other methods, such as hyperspectral imaging; however, their use is limited to the phenotyping of physiological plant traits [3]. Image compression techniques can be used to reduce the amount of data transmission required for remote analysis but have a non-negligible impact on the accuracy of image segmentation [55]. Therefore, a delicate balance must be struck between cost and

numerous factors that affect the overall precision, accuracy, and measurability of phenotypic traits [7].

2.4.3 Potential Advances For Low-Cost Automated Image-Based Plant Phenotyping

Low-cost plant phenotyping is not a nascent field of inquiry. However, many tracts of innovation have not been fully exploited. For instance, learning-based imaging techniques for analysis and processing can reduce analysis and segmentation costs but struggle when executed on resource-constrained hardware [28]. Therefore, there is a need for the development of efficient and optimised deep-learning or machine-learning models in resource-constrained environments. Similarly, many existing phenotyping systems are not generalizable to other plant species [3, 36] and rely on proprietary software or hardware [51]. Hence, there are opportunities to develop modifiable and flexible image-processing pipelines that are amenable to a broader variety of plant phenotyping contexts. Additionally, there are not many mobile applications for plant phenotyping. Mobile devices have advanced image sensors and can locally run image analysis and processing tasks, eliminating the need for cloud computing resources that increase phenotyping costs [7]. In short, numerous opportunities exist to improve the overall affordability and generalisability of plant phenotyping in low-cost settings.

2.5 Conclusion

Ensuring an adequate food supply in response to the growing human population is crucial [1], and image-based plant phenotyping has proven to be an effective tool in developing cultivars with maximum yield and resistance to various environmental stressors [11]. These methods are non-destructive, more time-efficient, replicable, and affordable than traditional methods [17]. However, image sensors in plant phenotyping generate large amounts of data, which necessitate robust image analysis and processing tools to handle this data effectively [8]. Large plant data throughput makes image-based plant phenotyping in automated greenhouse environments cost-prohibitive, as affordable off-the-shelf hardware and sensors often lack adequate local resources to store or process large amounts of data [53]. Consequently, remote cloud computing solutions handle image processing and analysis tasks [54], which can further increase operational costs in greenhouse management [4]. Image-data compression techniques, low-complexity image analysis and processing techniques, and low-cost off-the-shelf hardware are effective in reducing the cost of image-based plant phenotyping by reducing the amount of computation and data throughput required [55, 50, 4]. However, these solutions come with marked trade-offs in terms of accuracy, precision, depth, and breadth of plant trait analysis. Moreover, existing solutions are not generalizable to other environments and plant species [2, 3]. In short, there is a need for low-cost, open-source, flexible, modifiable, and robust plane

phenotyping image processing and analysis tools that are effective and efficient on resource-constrained hardware [4]. Future research should develop low-cost image processing tools that are precise, accurate, and adaptable to various plant monitoring contexts or parameters. In doing so, automated image-based techniques for plant monitoring will become more accessible in low-cost greenhouse environments.

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Bibliography

- [1] M. Carvajal-Yepes et al. “A Global Surveillance System for Crop Diseases”. In: *Science* 364.6447 (June 28, 2019), pp. 1237–1239. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.aaw1572. URL: <https://www.science.org/doi/10.1126/science.aaw1572> (visited on 05/21/2025).
- [2] Jayme Garcia Arnal Barbedo. “Digital Image Processing Techniques for Detecting, Quantifying and Classifying Plant Diseases”. In: *SpringerPlus* 2.1 (Dec. 7, 2013), pp. 1–12. ISSN: 2193-1801. DOI: 10.1186/2193-1801-2-660. URL: <https://doi.org/10.1186/2193-1801-2-660> (visited on 05/11/2025).
- [3] Lei Li, Qin Zhang, and Danfeng Huang. “A Review of Imaging Techniques for Plant Phenotyping”. In: *Sensors (Basel, Switzerland)* 14.11 (Oct. 24, 2014), pp. 20078–20111. ISSN: 1424-8220. DOI: 10.3390/s141120078. PMID: 25347588. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4279472/> (visited on 04/20/2025).
- [4] Massimo Minervini, Hanno Scharf, and Sotirios A. Tsaftaris. “Image Analysis: The New Bottleneck in Plant Phenotyping”. In: *IEEE Signal Processing Magazine* 32.4 (July 2015), pp. 126–131. ISSN: 1053-5888, 1558-0792. DOI: 10.1109/MSP.2015.2405111. URL: <https://ieeexplore.ieee.org/document/7123050/> (visited on 05/24/2025).
- [5] Huichun Zhang et al. “High-Throughput Phenotyping of Plant Leaf Morphological, Physiological, and Biochemical Traits on Multiple Scales Using Optical Sensing”. In: *The Crop Journal* 11.5 (Oct. 1, 2023), pp. 1303–1318. ISSN: 2214-5141. DOI: 10.1016/j.cj.2023.04.014. URL: <https://www.sciencedirect.com/science/article/pii/S2214514123000740> (visited on 05/20/2025).
- [6] Sruti Das Choudhury. “Segmentation Techniques and Challenges in Plant Phenotyping”. In: *Intelligent Image Analysis for Plant Phenotyping*. 1st ed. CRC Press, 2020, pp. 69–91. ISBN: 978-1-315-17730-4.
- [7] Mark Müller-Linow et al. “Plant Screen Mobile: An Open-Source Mobile Device App for Plant Trait Analysis”. In: *Plant Methods* 15.1 (Dec. 2019), pp. 1–11. ISSN: 1746-4811. DOI: 10.1186/s13007-019-0386-z. URL: <https://plantmethods.biomedcentral.com/articles/10.1186/s13007-019-0386-z> (visited on 05/25/2025).

- [8] Carla Gomes. “Computational Sustainability Computational Methods for a Sustainable Environment, Economy, and Society”. In: *The BRIDGE* 39.4 (2009), pp. 5–13. URL: <https://computational-sustainability.cis.cornell.edu/pdfs/Gomes-NAE-Bridge-Winter09.pdf> (visited on 05/25/2025).
- [9] Thomas J. Orton. “Introduction”. In: *Horticultural Plant Breeding*. Elsevier, 2020, pp. 3–7. ISBN: 978-0-12-815396-3. DOI: 10.1016/B978-0-12-815396-3.09986-8. URL: <https://linkinghub.elsevier.com/retrieve/pii/B9780128153963099868> (visited on 05/16/2025).
- [10] J.P. Grime. “Evidence for the Existence of Three Primary Strategies in Plants and Its Relevance to Ecological and Evolutionary Theory”. In: *The American Naturalist* 111.982 (1997), p. 1169. DOI: 10.1086/283244. URL: <https://www.journals.uchicago.edu/doi/abs/10.1086/283244> (visited on 05/19/2025).
- [11] Achala Anand, Madhumitha Subramanian, and Debasish Kar. “Breeding Techniques to Dispense Higher Genetic Gains”. In: *Frontiers in Plant Science* 13 (Jan. 19, 2023), pp. 1–5. ISSN: 1664-462X. DOI: 10.3389/fpls.2022.1076094. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2022.1076094/full> (visited on 05/16/2025).
- [12] Fabio Fiorani and Ulrich Schurr. “Future Scenarios for Plant Phenotyping”. In: *Annual Review of Plant Biology* 64.1 (Apr. 29, 2013), pp. 267–291. ISSN: 1543-5008, 1545-2123. DOI: 10.1146/annurev-arplant-050312-120137. URL: <https://www.annualreviews.org/doi/10.1146/annurev-arplant-050312-120137> (visited on 05/14/2025).
- [13] W. Johannsen. “The Genotype Conception of Heredity”. In: *The American Naturalist* 45.531 (Mar. 1911), pp. 129–159. ISSN: 0003-0147, 1537-5323. DOI: 10.1086/279202. URL: <https://www.journals.uchicago.edu/doi/10.1086/279202> (visited on 05/14/2025).
- [14] Qiaosheng Guo and Zaibiao Zhu. “Phenotyping of Plants”. In: *Encyclopedia of Analytical Chemistry*. John Wiley & Sons, Ltd, 2014, pp. 1–15. ISBN: 978-0-470-02731-8. DOI: 10.1002/9780470027318.a9934. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470027318.a9934> (visited on 05/14/2025).
- [15] Semira Ebrahim Mohammed. “Review on Plant Phenotyping”. In: *Agriculture and Biological Research* 1.41 (2025), pp. 1222–1225. URL: <https://www.abrinternationaljournal.org/abstract/review-on-plant-phenotyping-1102322.html> (visited on 05/05/2025).
- [16] Achim Walter, Frank Liebisch, and Andreas Hund. “Plant Phenotyping: From Bean Weighing to Image Analysis”. In: *Plant Methods* 11.1 (Mar. 4, 2015), pp. 1–8. ISSN: 1746-4811. DOI: 10.1186/s13007-015-0056-8. URL: <https://doi.org/10.1186/s13007-015-0056-8> (visited on 05/05/2025).

- [17] Sruti Das Choudhury, Ashok Samal, and Tala Awada. “Leveraging Image Analysis for High-Throughput Plant Phenotyping”. In: *Frontiers in Plant Science* 10 (Apr. 24, 2019), pp. 1–2. ISSN: 1664-462X. DOI: 10.3389/fpls.2019.00508. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2019.00508/full> (visited on 05/21/2025).
- [18] Juanjuan Zhang et al. “Leaf Area Index Estimation Model for UAV Image Hyperspectral Data Based on Wavelength Variable Selection and Machine Learning Methods”. In: *Plant Methods* 17.1 (May 3, 2021), pp. 1–2. ISSN: 1746-4811. DOI: 10.1186/s13007-021-00750-5. URL: <https://doi.org/10.1186/s13007-021-00750-5> (visited on 05/25/2025).
- [19] J. Praveen Kumar and S. Domnic. “Rosette Plant Segmentation with Leaf Count Using Orthogonal Transform and Deep Convolutional Neural Network”. In: *Machine Vision and Applications* 31.1–2 (Feb. 2020), pp. 1–2. ISSN: 0932-8092, 1432-1769. DOI: 10.1007/s00138-019-01056-2. URL: <http://link.springer.com/10.1007/s00138-019-01056-2> (visited on 05/25/2025).
- [20] Amin Afzal, Sjoerd W. Duiker, and John E. Watson. “Leaf Thickness to Predict Plant Water Status”. In: *Biosystems Engineering* 156 (Apr. 1, 2017), pp. 63–70. ISSN: 1537-5110. DOI: 10.1016/j.biosystemseng.2017.01.011. URL: <http://www.scopus.com/inward/record.url?scp=85013072400&partnerID=8YFLogxK> (visited on 05/25/2025).
- [21] Roselyne Ishimwe, K. Abutaleb, and F. Ahmed. “Applications of Thermal Imaging in Agriculture—A Review”. In: *Advances in Remote Sensing* 03.03 (2014), pp. 128–140. ISSN: 2169-267X, 2169-2688. DOI: 10.4236/ars.2014.33011. URL: <http://www.scirp.org/journal/doi.aspx?DOI=10.4236/ars.2014.33011> (visited on 05/19/2025).
- [22] Kenta Watanabe et al. “Fundamental Study on Water Stress Detection in Sugarcane Using Thermal Image Combined with Photosynthesis Measurement Under a Greenhouse Condition”. In: *Sugar Tech* 24.5 (Oct. 1, 2022), pp. 1382–1390. ISSN: 0974-0740. DOI: 10.1007/s12355-021-01087-y. URL: <https://doi.org/10.1007/s12355-021-01087-y> (visited on 05/25/2025).
- [23] Gernot Bodner et al. “Hyperspectral Imaging: A Novel Approach for Plant Root Phenotyping”. In: *Plant Methods* 14.1 (Oct. 3, 2018), pp. 1–2. ISSN: 1746-4811. DOI: 10.1186/s13007-018-0352-1. URL: <https://doi.org/10.1186/s13007-018-0352-1> (visited on 05/25/2025).
- [24] Sruti Das Choudhury et al. “Automated Stem Angle Determination for Temporal Plant Phenotyping Analysis”. In: *Proceedings of the IEEE International Conference on Computer Vision Workshops*. 2017, pp. 2022–2029. DOI: 10.1109/ICCVW.2017.237. URL: https://openaccess.thecvf.com/content_ICCV_2017_workshops/w29/html/Choudhury_Automated_Stem_Angle_ICCV_2017_paper.html (visited on 05/25/2025).

- [25] Chongyuan Zhang et al. “High-Throughput Phenotyping of Seed/Seedling Evaluation Using Digital Image Analysis”. In: *Agronomy* 8.5 (5 May 2018), pp. 1–2. ISSN: 2073-4395. DOI: 10.3390/agronomy8050063. URL: <https://www.mdpi.com/2073-4395/8/5/63> (visited on 05/25/2025).
- [26] Alebel Mekuriaw Abebe et al. “Image-Based High-Throughput Phenotyping in Horticultural Crops”. In: *Plants* 12.10 (10 Jan. 2023), pp. 1–16. ISSN: 2223-7747. DOI: 10.3390/plants12102061. URL: <https://www.mdpi.com/2223-7747/12/10/2061> (visited on 05/25/2025).
- [27] R. H. Laxman et al. “Non-Invasive Quantification of Tomato (*Solanum Lycopersicum* L.) Plant Biomass through Digital Imaging Using Phenomics Platform”. In: *Indian Journal of Plant Physiology* 23.2 (June 1, 2018), pp. 369–375. ISSN: 0974-0252. DOI: 10.1007/s40502-018-0374-8. URL: <https://doi.org/10.1007/s40502-018-0374-8> (visited on 05/24/2025).
- [28] Shona Nabwire et al. “Review: Application of Artificial Intelligence in Phenomics”. In: *Sensors* 21.13 (June 25, 2021), pp. 1–18. ISSN: 1424-8220. DOI: 10.3390/s21134363. URL: <https://www.mdpi.com/1424-8220/21/13/4363> (visited on 05/25/2025).
- [29] Ruicheng Qiu et al. “Sensors for Measuring Plant Phenotyping: A Review”. In: *International Journal of Agriculture and Biological Engineering* 11.2 (2018), pp. 1–14. DOI: <https://dx.doi.org/10.25165/j.ijabe.20181102.2696>. URL: <http://www.ijabe.net/cn/article/doi/10.25165/j.ijabe.20181102.2696> (visited on 05/09/2025).
- [30] Mahmood R Golzarian et al. “Accurate Inference of Shoot Biomass from High-Throughput Images of Cereal Plants”. In: *Plant Methods* 7.1 (Dec. 2011), pp. 1–2. ISSN: 1746-4811. DOI: 10.1186/1746-4811-7-2. URL: <https://plantmethods.biomedcentral.com/articles/10.1186/1746-4811-7-2> (visited on 05/26/2025).
- [31] V. Hoyos-Villegas et al. “Ground-Based Digital Imaging as a Tool to Assess Soybean Growth and Yield”. In: *Crop Science* 54.4 (2014), pp. 1756–1768. ISSN: 1435-0653. DOI: 10.2135/cropsci2013.08.0540. URL: <https://onlinelibrary.wiley.com/doi/abs/10.2135/cropsci2013.08.0540> (visited on 05/26/2025).
- [32] Rijad Sarić et al. “Applications of Hyperspectral Imaging in Plant Phenotyping”. In: *Trends in Plant Science* 27.3 (Mar. 1, 2022), pp. 301–315. ISSN: 1360-1385. DOI: 10.1016/j.tplants.2021.12.003. URL: <https://www.sciencedirect.com/science/article/pii/S1360138521003460> (visited on 05/22/2025).
- [33] Hartmut K. Lichtenthaler and Ursula Rinderle. “The Role of Chlorophyll Fluorescence in The Detection of Stress Conditions in Plants”. In: *C R C Critical Reviews in Analytical Chemistry* 19 (Supplement 1 Jan. 1, 1988), pp. 30–79. ISSN: 0007-8980. DOI: 10.1080/15476510.1988.10401466. URL: <https://doi.org/10.1080/15476510.1988.10401466> (visited on 05/22/2025).

- [34] Philip J. Swarbrick, Paul Schulze-Lefert, and Julie D. Scholes. “Metabolic Consequences of Susceptibility and Resistance (Race-Specific and Broad-Spectrum) in Barley Leaves Challenged with Powdery Mildew”. In: *Plant, Cell & Environment* 29.6 (2006), pp. 1061–1076. ISSN: 1365-3040. DOI: 10.1111/j.1365-3040.2005.01472.x. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-3040.2005.01472.x> (visited on 05/26/2025).
- [35] Carel W. Windt et al. “MRI of Long-Distance Water Transport: A Comparison of the Phloem and Xylem Flow Characteristics and Dynamics in Poplar, Castor Bean, Tomato and Tobacco”. In: *Plant, Cell & Environment* 29.9 (2006), pp. 1715–1729. ISSN: 1365-3040. DOI: 10.1111/j.1365-3040.2006.01544.x. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-3040.2006.01544.x> (visited on 05/26/2025).
- [36] Samuel Arvidsson, Paulino Pérez-Rodríguez, and Bernd Mueller-Roeber. “A Growth Phenotyping Pipeline for Arabidopsis Thaliana Integrating Image Analysis and Rosette Area Modeling for Robust Quantification of Genotype Effects”. In: *New Phytologist* 191.3 (2011), pp. 895–907. ISSN: 1469-8137. DOI: 10.1111/j.1469-8137.2011.03756.x. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1469-8137.2011.03756.x> (visited on 05/26/2025).
- [37] Sruti Das Choudhury et al. “Holistic and Component Plant Phenotyping Using Temporal Image Sequence”. In: *Plant Methods* 14.1 (May 10, 2018), pp. 1–4. ISSN: 1746-4811. DOI: 10.1186/s13007-018-0303-x. URL: <https://doi.org/10.1186/s13007-018-0303-x> (visited on 05/26/2025).
- [38] Esmael Hamuda, Martin Glavin, and Edward Jones. “A Survey of Image Processing Techniques for Plant Extraction and Segmentation in the Field”. In: *Computers and Electronics in Agriculture* 125 (July 2016), pp. 184–199. ISSN: 01681699. DOI: 10.1016/j.compag.2016.04.024. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0168169916301557> (visited on 05/26/2025).
- [39] Zhibin Wang et al. “Image Segmentation of Overlapping Leaves Based on Chan–Vese Model and Sobel Operator”. In: *Information Processing in Agriculture* 5.1 (Mar. 1, 2018), pp. 1–10. ISSN: 2214-3173. DOI: 10.1016/j.inpa.2017.09.005. URL: <https://www.sciencedirect.com/science/article/pii/S2214317317301270> (visited on 05/26/2025).
- [40] Suraj Gampa. “A Data-driven Approach for Detecting Stress in Plants Using Hyperspectral Imagery”. University of Nebraska-Lincoln, 2019.
- [41] Yuhao Chen et al. “Leaf Segmentation by Functional Modeling”. In: *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). Long Beach, CA, USA: IEEE, June 2019, pp. 1–10. ISBN: 978-1-7281-2506-0. DOI: 10.1109/CVPRW.2019.

00326. URL: <https://ieeexplore.ieee.org/document/9025489/> (visited on 05/26/2025).
- [42] J. Praveen Kumar and S. Domnic. “Image Based Leaf Segmentation and Counting in Rosette Plants”. In: *Information Processing in Agriculture* 6.2 (June 2019), pp. 233–246. ISSN: 22143173. DOI: 10.1016/j.inpa.2018.09.005. URL: <https://linkinghub.elsevier.com/retrieve/pii/S2214317318301562> (visited on 05/26/2025).
 - [43] Jean-Michel Pape and Christian Klukas. “Utilizing Machine Learning Approaches to Improve the Prediction of Leaf Counts and Individual Leaf Segmentation of Rosette Plant Images”. In: *Proceedings of the Proceedings of the Computer Vision Problems in Plant Phenotyping Workshop 2015*. Proceedings of the Computer Vision Problems in Plant Phenotyping Workshop 2015. Swansea: British Machine Vision Association, 2015, pp. 1–2. ISBN: 978-1-901725-55-1. DOI: 10.5244/C.29.CVPPP.3. URL: <http://www.bmva.org/bmvc/2015/cvppp/papers/paper003/index.html> (visited on 05/26/2025).
 - [44] Alwaseela Abdalla et al. “Fine-Tuning Convolutional Neural Network with Transfer Learning for Semantic Segmentation of Ground-Level Oilseed Rape Images in a Field with High Weed Pressure”. In: *Computers and Electronics in Agriculture* 167 (Dec. 2019), pp. 1–2. ISSN: 01681699. DOI: 10.1016/j.compag.2019.105091. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0168169919317107> (visited on 05/26/2025).
 - [45] Amine Saddik et al. “Mapping Agricultural Soil in Greenhouse Using an Autonomous Low-Cost Robot and Precise Monitoring”. In: *Sustainability* 14.23 (23 Jan. 2022), pp. 1–24. ISSN: 2071-1050. DOI: 10.3390/su142315539. URL: <https://www.mdpi.com/2071-1050/14/23/15539> (visited on 05/24/2025).
 - [46] Brooke Bruning et al. “The Development of Hyperspectral Distribution Maps to Predict the Content and Distribution of Nitrogen and Water in Wheat (*Triticum Aestivum*)”. In: *Frontiers in Plant Science* 10 (Oct. 30, 2019), pp. 1–2. ISSN: 1664-462X. DOI: 10.3389/fpls.2019.01380. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2019.01380/full> (visited on 05/26/2025).
 - [47] Laura S. Peirone et al. “Assessing the Efficiency of Phenotyping Early Traits in a Greenhouse Automated Platform for Predicting Drought Tolerance of Soybean in the Field”. In: *Frontiers in Plant Science* 9 (May 3, 2018), pp. 1–2. ISSN: 1664-462X. DOI: 10.3389/fpls.2018.00587. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2018.00587/full> (visited on 05/26/2025).
 - [48] Daoliang Li et al. “High-Throughput Plant Phenotyping Platform (HT3P) as a Novel Tool for Estimating Agronomic Traits From the Lab to the Field”. In: *Frontiers in Bioengineering and Biotechnology* 8 (Jan. 13, 2021), pp. 2–7. ISSN: 2296-4185. DOI: 10.3389/fbioe.2020.623705.

URL: <https://www.frontiersin.org/articles/10.3389/fbioe.2020.623705/full> (visited on 05/24/2025).

- [49] Max R. Lien et al. “A Low-Cost and Open-Source Platform for Automated Imaging”. In: *Plant Methods* 15.1 (Dec. 2019), pp. 2–13. ISSN: 1746-4811. DOI: 10.1186/s13007-019-0392-1. URL: <https://plantmethods.biomedcentral.com/articles/10.1186/s13007-019-0392-1> (visited on 05/24/2025).
- [50] Jose C. Tovar et al. “Raspberry Pi-Powered Imaging for Plant Phenotyping”. In: *Applications in Plant Sciences* 6.3 (2018), pp. 1–2. ISSN: 2168-0450. DOI: 10.1002/aps3.1031. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/aps3.1031> (visited on 05/24/2025).
- [51] Angelika Czedik-Eysenberg et al. “The ‘PhenoBox’, a Flexible, Automated, Open-Source Plant Phenotyping Solution”. In: *New Phytologist* 219.2 (2018), pp. 808–823. ISSN: 1469-8137. DOI: 10.1111/nph.15129. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/nph.15129> (visited on 05/26/2025).
- [52] Dawei Li et al. “Digitization and Visualization of Greenhouse Tomato Plants in Indoor Environments”. In: *Sensors* 15.2 (2 Feb. 2015), pp. 4019–4051. ISSN: 1424-8220. DOI: 10.3390/s150204019. URL: <https://www.mdpi.com/1424-8220/15/2/4019> (visited on 05/24/2025).
- [53] Christian Bauckhage and Kristian Kersting. “Data Mining and Pattern Recognition in Agriculture”. In: *KI - Künstliche Intelligenz* 27.4 (Nov. 1, 2013), pp. 1–11. ISSN: 1610-1987. DOI: 10.1007/s13218-013-0273-0. URL: <https://doi.org/10.1007/s13218-013-0273-0> (visited on 05/25/2025).
- [54] Olivier Debauche et al. “Cloud Architecture for Plant Phenotyping Research”. In: *Concurrency and Computation: Practice and Experience* 32.17 (Sept. 10, 2020), pp. 1–2. ISSN: 1532-0626, 1532-0634. DOI: 10.1002/cpe.5661. URL: <https://onlinelibrary.wiley.com/doi/10.1002/cpe.5661> (visited on 05/26/2025).
- [55] Massimo Minervini and Sotirios A. Tsaftaris. “Application-Aware Image Compression for Low Cost and Distributed Plant Phenotyping”. In: 2013 18th International Conference on Digital Signal Processing (DSP). July 2013, pp. 1–6. DOI: 10.1109/ICDSP.2013.6622670. URL: <https://ieeexplore.ieee.org/abstract/document/6622670> (visited on 05/31/2025).